

Glycosylated IgG antibodies accelerated the recovery of haemorrhagic fever with renal syndrome patients

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eLife Assessment

The authors investigated the potential role of IgG N-glycosylation in Haemorrhagic Fever with Renal Syndrome (HFRS), which may offer significant insights for understanding molecular mechanisms and for the development of therapeutic strategies for this infectious disease. The findings are **useful** to the field, although the strength of evidence to support the findings is **incomplete**. Several issues need to be addressed, including more detail on the background, methods, and results. Additional statistical tests should be performed, and the conclusions should reflect the correlational findings of the paper.

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Abstract

Haemorrhagic fever with renal syndrome (HFRS) is a fatal disease caused by Hantaan virus (HTNV) infection. Humoral immunity is essential for effective viral clearance, but the glycosylation characteristics of immunoglobulin G (IgG) in HFRS patients is not well known. Peripheral blood mononuclear cells from HFRS patients were obtained for B subset analysis using scRNA-seq and flow cytometry. HTNV specific IgG antibody titers were detected by enzyme linked immunosorbent assay, and IgG glycosylation was analyzed by ultra-

performance liquid chromatography. The proportions of the antibody-secreting memory (ASM) B cells and plasmablasts (PB) were significantly expanded among acute HFRS patients. We discovered significantly increased fucosylated IgG and decreased bisecting N-acetylglucosamine during the convalescent phase of HTNV infection. However, positive correlations were observed between ASM subsets and galactosylation/sialylation in the IgG Fc region, and between PB subsets and sialylation. Notably, the glycosylation-related genes were primarily differentially expressed in the ASM and PB subclusters, such as *PRN1* and *PRN2*, which were enriched in the N-glycosylation modifications of proteins through asparagine. Our findings indicated that IgG N-glycosylation may play a crucial role in combating HTNV infection and contributing to clinical recovery, which provided new insights for optimizing glycoengineered therapeutic antibodies.

Introduction

The causative agent of Haemorrhagic fever with renal syndrome (HFRS) is Hantaan virus, which belongs to the family *Hantaviridae* and is primarily transmitted by rodents (Zheng et al., 2019). China is the most heavily affected country by HFRS, accounting for >90% of global cases, and a total of 32,462 cases of HFRS were reported in China during 2019–2022, with a fatality rate ranging from 0.1 to 15% (R. Li et al., 2023) (<https://www.chinacdc.cn>).

Immunoglobulin G (IgG) is one of the best-understood glycoproteins and each constant heavy 2 domain (CH2) of the fragment crystallisable (Fc) region carries a covalently attached biantennary N-glycan at the highly conserved asparagine at residue 297 (Trastoy et al., 2018). Functionally, minor alterations in the glycan composition can drastically affect the structural conformation of the Fc region with enormous IgG effector functions (Abu-Raya, Michalski, Sadarangani, & Lavoie, 2020). IgG modulates both pro- and anti-inflammatory activities depending on the moieties of the glycans. For example, sialylation of IgG-Fc reverses a pro-inflammatory process into an anti-inflammatory effect (Pereira et al., 2020). It has been evidenced that progression of dengue virus and severe acute respiratory syndrome coronavirus 2 (SARS-CoV-2) infection is regulated by specific IgG glycosylation (Hou et al., 2021; Wang et al., 2017). The human immune response plays a fundamental role in anti-viral infections. Massive and rapidly expanding plasmablast (PB) cells are required for the recovery of HFRS patients (García et al., 2017; Y. Li et al., 2020), however, the increased amount of circulating CD27⁺IgD⁺ B cells may partially associate with kidney dysfunction (Kerkman et al., 2021). However, there is limited report regarding the regulation of IgG glycosylation in HFRS.

In the present study, we combined scRNA-seq and flow cytometry to reveal the phenotypes of B cell responses during Hantaan virus (HTNV) infection, and explored the transcriptomic features of reactive B cell subsets. A total of 166 HFRS patients, including 65 paired HFRS samples, were involved to profile the IgG-Fc glycosylation and investigate the potential regulatory role of IgG-Fc glycosylation in HFRS pathogenesis.

Results

B cell compositional characteristics in HFRS patients

To characterize the humoral immune profiles in HFRS patients, we enrolled suspected 166 patients who were admitted to Baoji Central Hospital in Shaanxi Province, China, between October 2019 and January 2022, of whom 65 met the inclusion criteria and were included in the study (Figure 1). We identified a total of eight cell subpopulations, including subsets of CD4⁺ T cells, CD8⁺ T cells, CD14⁺ monocytes, NK cells, B cells, platelet, endothelial progenitor cells, and red blood cells (Figure

2—figure supplement 1A and B). The proportions of these cell types were comparable between acute HFRS patients and healthy controls, with no significant batch effects observed (**Figure 2—figure supplement 1C**). CD4 T cells (31.4%) were the predominant subset prior to HTNV infection, whereas CD8 T cells (44%) became the dominant population post-infection. Notably, the proportions of NK cells (15.9% to 5.6%, $p < 0.001$) and platelets (3.1% to 0.6%, $p < 0.01$) significantly decreased, while the level of CD14 monocytes (15.9% to 32.3%, $p < 0.05$) nearly doubled. However, the overall proportions of B cells did not exhibit significant changes (**Figure 2—figure supplement 1D and E**).

To precisely elucidate the role of B cell subsets in anti-HTNV infection, we classified B cells into eight subsets, including antibody-secreting memory B cells (ASM), double-negative B cells (DN), intermediate memory B cells (IM), marginal zone-like B cells (MZB), naïve B cells (naïve B), PB, quiescent resting memory B cells (RM), and exhausted tissue-like memory B cells (TLM) (**Figure 2A and B**, and **Figure 2—figure supplement 1F**). We discovered that significant expansion of ASM, PB, and RM cell populations in acute HFRS patients compared to healthy controls, accompanied by contraction of DN, IM, and naïve B cell compartments post-infection (**Figure 2C**). In healthy controls, naïve B cells (42.87%), DN B cells (26.65%), and TLM B cells (16.13%) constituted the majority of the B cell population. However, ASM (30.96%), naïve B cells (23.85%), and PB cells (11.12%) became the dominant subsets after HTNV infection (**Figure 2D** and Supplementary file 1).

To validate the accuracy of our single-cell subpopulation analysis, we employed flow cytometry to quantify the B cell subsets in acute and convalescent HFRS patients (**Figure 2E**). The results demonstrated that ASM (39.48% vs. 24.57%, $p < 0.001$) and PB (23.97% vs. 2.61%, $p < 0.001$) subsets were significantly amplified in the acute phase compared to the convalescent phase (**Figure 2F**). Conversely, the number of naïve B cells (32.78% vs. 48.92%, $p = 0.003$) and CSM cells (18.65% vs. 27.15%, $p = 0.008$) rapidly recovered in the convalescent stage (**Figure 2F**), consistent with the observations at the transcriptome level. No significant differences were observed for other B cell subtypes between the acute and convalescent stages (**Figure 2—figure supplement 2**). These results clearly exhibited that several B cell subpopulations were effectively activated following HTNV infection.

Dynamic characteristics of the IgG glycome in HFRS patients

HFRS patients were further stratified into three age groups (≤ 44 , 45–59, and ≥ 60 years). IgG glycosylation was analyzed by ultra-performance liquid chromatography. As shown in Supplementary file 2, no significant differences in IgG glycosylation were observed between age groups, except for galactosylation. Similarly, no gender-based differences in IgG-Fc glycosylation were noted (Supplementary file 3). However, the levels of IgG with bisecting GlcNAc (15.72% vs. 14.51%, $p < 0.001$), galactosylated IgG (74.26% vs. 71.15%, $p < 0.001$), and sialylated IgG (22.44% vs. 21.62%, $p < 0.001$) were significantly higher in the acute phase compared to the convalescent phase. Conversely, fucosylated IgG levels (94.29% vs. 94.89%, $p < 0.001$) were lower in the acute phase (Supplementary file 4), indicating that IgG N-glycosylation plays a critical role in the anti-infection process.

Among the 130 paired blood samples from 65 patients, 38 (29.2%) were negative for HTNV nucleocapsid protein-specific IgG antibodies, while 92 (70.8%) were seropositive. For the 24 patients with both seronegative and seropositive samples, levels of bisecting GlcNAc (16.16% vs. 14.54%, $p < 0.001$), galactosylation (74.12% vs. 70.94%, $p = 0.006$), and sialylation (22.92% vs. 21.80%, $p = 0.008$) were higher in the seronegative period, while that of fucosylation was lower (94.36% vs. 94.74%, $p = 0.024$) in this period (**Figure 3A**). Notably, bisecting GlcNAc and sialylated IgG levels decreased as HTNV-NP-specific antibody titers increased (**Figure 3B**), highlighting a potential link between antibody levels and IgG-Fc glycosylation.

Figure 1.

The flow chart of HFRS patient recruitment in Baoji Central Hospital in Shaanxi province from November 2019 to January 2022.

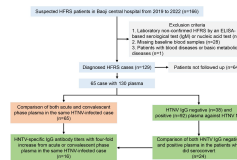


Figure 2.

Immunophenotypic remodeling in the B-cell subsets during acute HFRS.

(A) t-distributed stochastic neighbor embedding (t-SNE) plot showing antibody secreting memory B cells (ASM), double negative (DN) B cells, intermediate memory cells (IM), marginal zone-like cells (MZB), naïve B cells, plasmablasts (PB), quiescent resting memory cells (RM), and exhausted tissue-like memory B cells (TLM) of PBMCs identified using an integrated and classification analysis. (B) t-SNE projection of canonical markers, including CD19, CD27, CD38, IgD, and IgM. (C) Proportions of the eight B cell subsets, colored by the healthy group (green) and HTNV groups (red). Boxplot features: minimum box, 25th percentile; center, median; maximum box, 75th percentile. (D) Frequency of the eight B cell subsets in between the healthy group (n = 8) and HTNV groups (n = 15). (E) FACS gating strategy for the measurement of the B-cell subsets: activated memory B cells (CD21⁺ CD27⁺, AM), RM B cells (CD21⁺ CD27⁻), IM B cells (CD21⁺ CD27⁻), TLM B cells (CD21⁺ CD27⁻), naïve B cells (CD27⁺ IgD⁺), MZB B cells (CD27⁺ IgD⁺), ASM B cells (CD27⁺ IgD⁻), DN B cells (CD27⁻ IgD⁻), PB (CD38⁺ CD27⁺), and class-switched memory B cells (CD38⁺ CD27⁺, CSM), respectively. (F) The proportion of ASM, CSM, naïve B, and PB cells in CD19⁺ B cells in the acute and convalescent HFRS patients. Data are presented as mean ± SEM in panels C and F. *p<0.05 (Student's t-test).

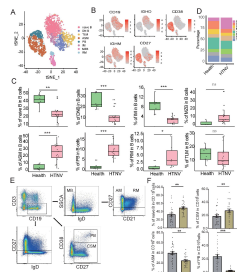
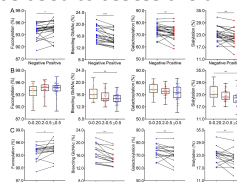


Figure 3.

Glycosylation modification of antibodies associated with HTNV infection.

(A) Changes of different glycosylation types in both HTNV-NP specific IgG negative and positive plasma from 24 HFRS patients. (B) Differential IgG glycosylation patterns across antibody titer levels quantified by ELISA. (C) Comparison of different glycosylation levels before and after the fourfold increase in the IgG antibody titers. *p<0.05, **p<0.01, ***p<0.001. For paired samples, Wilcoxon Signed Rank Test was used to assess the difference.



The IgG glycome in HFRS patients with seroconversion

A 4-fold or greater increase in HTNV-NP-specific antibody titers usually indicates a protective humoral immune response during the acute phase. We observed a significant increase in the Fc fucosylation (94.18% vs. 94.48%, $p=0.044$) and a decrease in bisecting GlcNAc (16.41% vs. 14.71%, $p=0.001$) post the 4-fold antibody titer increase compared to the baseline. However, no significant differences were observed in galactosylation and sialylation (**Figure 3C**), suggesting that fucosylation and bisecting GlcNAc may play key roles in anti-HTNV infection and recovery.

The potential source of the IgG glycome during HTNV infection

Positive correlations were between the ASM subsets and galactosylation/sialylation in the antibody Fc region, and between the PB subsets and sialylation (**Figure 4A-C**). Therefore, we speculated that galactosylated antibodies may primarily originate from the ASM subsets, while sialylated antibodies may derive from the ASM and PB subsets.

To investigate the process of glycosylated antibody production secreted by the ASM and PB subsets, we identified DEGs among the eight B cell subpopulations. Notably, the most DEGs were observed in the ASM (470), naïve (312), and TLM (333) B subpopulations, respectively (**Figure 4—figure supplement 1A** and Supplementary file 5). Specifically, among all the upregulated genes, the expressions of glycosylation-related, mitochondrial respiratory-related, and peptide chain elongation-related genes among the ASM, PB, and RM populations were drastically elevated after HTNV infection, including *RPN2*, *MT-ND6*, and *RPS26* (**Figure 4—figure supplement 1B** and Supplementary file 6). On the contrary, the upregulated genes in the naïve, DN, and IM populations were related to anti-inflammatory and immunosuppressive effects, inhibition of the antioxidative function, response to environmental stress, and B-cell growth and development, such as *TSC22D3*, *TXNIP*, *DUSP1*, and *KLF6*, whereas the downregulated genes played an important role in the inflammatory response, cell motility, and antigen presentation, including *LTB*, *ACTG1*, and *HLA-DRB5* (**Figure 4—figure supplement 1C** and Supplementary file 7).

Unexpectedly, we observed significant changes in the expression of glycosylation-related genes (GRGs) in the ASM, PB, and RM subclusters during HTNV infection compared to healthy controls. *PRN1* and *PRN2* were prominently upregulated, which encoded subunits of the oligosaccharyltransferase complex (**Figure 4D**). This implied that the process of glycosylation during HTNV infection was rapidly activated among the ASM and PB subclusters. In the ASM subcluster, the fold change of GRGs from all four glycosylation modification-related genes, aside from *PRN1* and *PRN2*, was similar, suggesting that ASM may undergo complex and diverse glycosylation modification processes.

To enhance our understanding of the transcriptomic characteristics of GRG-high-expressing PB and ASM subclusters, we conducted a comprehensive pathway enrichment analysis and identified significant enrichment of pathways related to glycosylation modifications, including the endoplasmic reticulum protein-containing complex, rough endoplasmic reticulum, and protein N-linked glycosylation via asparagine (**Figure 4E**). Additionally, gene sets associated with glycosylation modifications showed positive regulation of these pathways ($p=0.002$, **Figure 4F**). These results indicated that glycosylation modifications in the ASM, PB, and RM subpopulations were dynamically regulated during the acute phase of HFRS, likely to satisfy the amplified anti-HTNV antibody requirement and accelerate viral clearance.

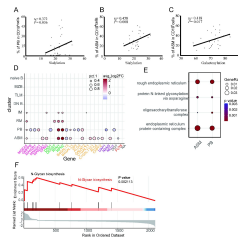


Figure 4.

Glycosylation modifications of antibodies primarily deprived from antibody-secreting and plasmablast subpopulations.

(A) The correlation between the proportion of PB cells and the sialylation level. (B) The correlation between the proportion of ASM cells and the sialylation level. (C) The correlation between the proportion of ASM cells and the galactosylation level. (D) Dot plot shows the expression levels of glycosylation-related genes in the eight B cell groups. The pink color represents mannose related genes, the green color represents N-glycosylation related genes, the orange color represents galactosylation related genes, the purple represents sialylation related genes, and the red color represents fucosylation related genes. (E) Enriched pathways in the plasmablast and ASM subsets by GO enrichment analysis of the DEGs. (F) The GSEA map presents the enrichment score of GRGs in N-Glycan biosynthesis.

Discussion

IgG glycosylation is a key regulator of inflammatory responses for infectious diseases, acting as a molecular switch between anti-inflammatory and pro-inflammatory effects upon antigenic challenge. One of the most significant findings of this study is the elevation in IgG fucosylation and reduction in bisecting GlcNAc levels at the time of a 4-fold increase in the HTNV-NP-specific antibody titers. Afucosylation is known to enhance antibody-dependent cellular cytotoxicity (ADCC) by increasing the affinity of IgG-Fc for the FcγRIIIa receptor on natural killer cells, leading to heightened production of inflammatory cytokines such as IL-1β, IL-6, IFN-γ, and TNF-α (Larsen et al., 2021 [↗](#); Lu, Suscovich, Fortune, & Alter, 2018 [↗](#); Wiecek, Braicu, Oliveira-Ferrer, Sehouli, & Blanchard, 2020 [↗](#)). Increase in bisecting GlcNAc linked to Fc induces enhanced affinity for the FcγRIIIa receptor and enhanced ADCC function of the antibodies (Giron et al., 2020 [↗](#); Hou et al., 2021 [↗](#); Irvine & Alter, 2020 [↗](#)). However, rather than directly influencing the Fc receptor binding, this modification enhances antibody functionality through afucosylation, which subsequently increases the binding affinity to the FcγRIIIa receptor (Ackerman et al., 2018 [↗](#); Jennewein & Alter, 2017 [↗](#)). Our findings provide compelling evidence that the dynamic alterations in glycosylation patterns, characterized by increased fucosylation and decreased bisecting GlcNAc levels during viral infection, are critically involved in disease pathogenesis and clinical recovery, suggesting their dual utility as both diagnostic biomarkers and promising therapeutic targets.

We also observed elevated galactosylation levels in the acute phase compared to the convalescent phase, a phenomenon also reported in SARS-CoV-2 infections but not in chronic diseases and cancers (Hou et al., 2021 [↗](#); D. Liu et al., 2018 [↗](#)). Increased galactosylation may expand the distance between CH2 domains, exposing key amino acids for FcγRIIIa binding and enhancing ADCC functions (Houde, Peng, Berkowitz, & Engen, 2010 [↗](#)). Simultaneously, the increase in the ASM subset, which positively correlated with galactosylation, may reflect polyclonal activation of memory B cells driven by pattern recognition receptors and cytokines, leading to the production of galactosylated antibodies. Similarly, sialylation levels were higher in the acute patients, potentially linked to the expansion of the ASM and PB subsets. In this study, the PB subset was significantly elevated in the acute phase and positively correlated with sialylation, suggesting that rapidly proliferating plasmablasts may contribute to the secretion of sialylated antibodies. However, the mechanisms behind this require further investigation.

Our findings demonstrate that the IgG-Fc glycosylation plays a pivotal role in HTNV infection and rapid recovery. Fucosylated oligosaccharides and the absence of bisecting GlcNAc are associated with favorable clinical outcomes. Importantly, specific B cell subsets, particularly ASMs and PBs, upregulated the secretion of galactosylated and sialylated antibodies, respectively. These findings would deepen our understanding of antibody glycosylation in anti-virus infection and provide a foundation for future studies aimed at optimizing glycoengineered therapeutic antibodies.

Materials and Methods

Study participants

Clinical specimens were collected from HFRS patients who were hospitalized in Baoji Central Hospital between October 2019 and January 2022. The HFRS patients were categorized into four clinical subtypes and clinical courses based on the previous reports (Ma et al., 2015 [↗](#); Tang et al., 2019 [↗](#); Zhang et al., 2022 [↗](#)). This study was approved by the ethics committee of the Shandong First Medical University & Shandong Academy of Medical Sciences (R201937). Written informed consent was obtained from each participant or their guardians.

Single-Cell RNA Sequencing and public database

Flow cytometry antibody staining was conducted as previously described (Chakraborty et al., 2022 [↗](#)). Dead cell and red blood cell were removed using the BD FACSAria™ III cell sorter, and PBMCs were collected for single-cell sequencing. The library construction was conducted as previously described (Jin et al., 2024 [↗](#)).

We obtained single-cell transcriptome sequencing data from the Gene Expression Omnibus (GEO) database (GSE161354), containing six HFRS patients and two healthy volunteers, as well as healthy control samples from the Genome Sequence Archive (GSA) database (HRA000203), matched with patient demographic information.

Single-Cell Data Analysis

The FastQC software was used to evaluate the data obtained to ensure the quality of the raw sequencing data. The raw data were mapped to the human reference genome (GRCh38, <https://cf.10xgenomics.com/supp/cell-exp/refdata-gex-GRCh38-2020-A.tar.gz> [↗](#)) using Cell Ranger. The Seurat R package (Version 4.4.0) was utilized for the merging and clustering of single-cell data. The Harmony R package and the anchor module of Seurat were used to remove batch effects between samples and groups for cell clustering. t-distributed Stochastic Neighbor Embedding (t-SNE) and Uniform Manifold Approximation and Projection (UMAP) were employed for dimensionality reduction and visualization of individual cells.

Cell Type Annotation

UMAP and t-SNE were used to reduce the dimension of all cells and to cluster them in a two-dimensional space based on shared features. The Single R (Version 2.2.0) package was used to independently infer the cell origin of each single cell based on a reference transcriptome dataset of pure cell types, for unbiased cell type identification from single-cell RNA sequencing data. Subsequently, specific highly expressed genes were combined with literature review for manual annotation to ultimately determine cell types. Classic biomarkers for specific cell types were used to identify cells in different clusters.

Differential Gene Identification and Functional Analysis

The *FindMarkers()* function in the Seurat package was used to identify differentially expressed genes (DEGs) between different cell groups, with criteria of $|\log_2FC| > 0.25$ and $p\text{-value} < 0.05$, $pct.1 > 0.25$. The ClusterProfiler R package was utilized for GO, KEGG, and GSEA enrichment analyses, and the ggplot2 and enrichplot R packages were used for visualization.

Measurement of HTNV specific antibodies

Levels of serum IgM and IgG antibodies to HTNV nucleocapsid protein (NP) were assessed through the enzyme-linked immunosorbent assay (ELISA) kit (Wantai BioPharm, Beijing, China) in accordance with the manufacturer's instructions.

Analysis of IgG glycans

The process of isolating, labelling, purifying, and analyzing plasma IgG was performed using previously established methods (Hou et al., 2019 [↗](#); P. Liu et al., 2023 [↗](#)).

Virus-specific B-cell detection

Virus-specific B-cell against HTNV were performed as previously described (Liechti & Roederer, 2019 [↗](#); Song et al., 2018 [↗](#)).

Statistical analysis

Categorical variables were compared using chi-square tests. Normality of numeric variables was assessed with the Kolmogorov-Smirnov test; non-normally distributed continuous variables were analyzed using non-parametric tests. Statistical analyses were performed with R software version 4.1.1 (R Core Team, New Zealand) and GraphPad Prism 8 (GraphPad Software, San Diego, CA), with a significance level of $\alpha = 0.05$ (two-tailed).

Supplementary materials

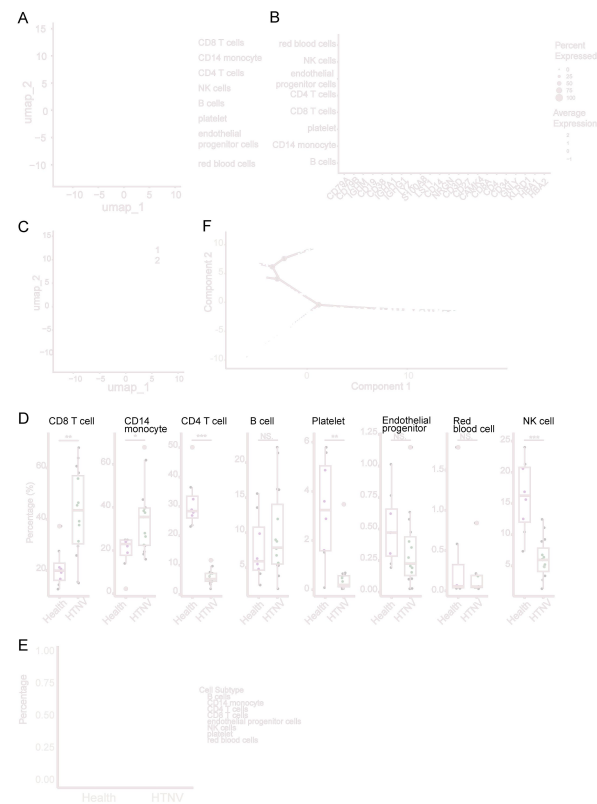


Figure 2—figure supplement 1.

Single-cell transcriptomes of PBMCs from patients with HFRS.

(A) Uniform manifold approximation and projection (UMAP) presentation of the major peripheral immune cell types in the PBMCs from 15 HFRS patients and 8 healthy controls. (B) Dot plot shows the expression levels of canonical marker-related genes in the different subpopulations. (C) UMAP presentation of the major peripheral immune cell types among HFRS and healthy control groups. (D) Proportions of the eight cell subsets, colored by the healthy group (green) and HTNV groups (red). Boxplot features: minimum box, 25th percentile; center, median; maximum box, 75th percentile. (E) Frequency of the eight cell subsets in between the healthy group (n = 8) and the HTNV group (n = 15). (F) The UMAP plot shows the differentiation trajectories of different cell types calculated by pseudotime analysis.

Dynamic analysis of the B-cell subsets in HFRS patients.

A

% of MZB in CD19⁺ cells

Acute Convalescent

ns

B

% of DN in CD19⁺ cells

Acute Convalescent

ns

C

% of TLM in CD19⁺ cells

Acute Convalescent

ns

D

% of AM in CD19⁺ cells

Acute Convalescent

ns

E

% of RM in CD19⁺ cells

Acute Convalescent

ns

F

% of IM in CD19⁺ cells

Acute Convalescent

ns

The differentially expressed genes and their functional changes in the B cell subsets post HTNV infection.

[illegible]

Additional information

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Data availability

The data underlying this article will be shared upon request to the corresponding author. The following dataset was generated:

Author(s)	Year	Dataset title	Dataset URL	Database and Identifier
Chuansong Quan	2025	Glycosylated IgG antibodies accelerated the recovery of haemorrhagic fever with renal syndrome patients	https://www.ncbi.nlm.nih.gov/sra/PRJNA1238574	Sequence Read Archive, PRINA1238574

Zhang H, Wang Y (2022) **Increased CD4⁺CD8⁺ double positive T cells during Hantaan virus infection** ID GSE161354. In the public domain at GEO (<https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE161354>).

Zheng Y, Liu X (2020) **A human circulating immune cell landscape in aging and COVID-19** ID HRA000203. In the public domain at GSA (<https://ngdc.cncb.ac.cn/gsa-human/browse/HRA000203>).

Author contributions

Haifeng Hou and Weifeng Shi designed and supervised this study. Chuansong Quan, Lu Wang, Jiming Gao, Yaoni Li, Houqiang Li, Hongzhi Liu, Qian Li, and Liqiong Zhao collected the samples and performed experiments. Chuansong Quan, Lu Wang, Jiming Gao, Xiaoyu Xu, Zixuan Gao, Wenxu Ruan, and Weijia Xing analyzed the data and performed the statistical analyses. Lu Wang, Chuansong Quan, Jiming Gao, and Xiaoyu Xu wrote the manuscript. Haifeng Hou, Weifeng Shi, Weijia Xing, and Michael J. Carr reviewed and edited the manuscript. All authors had full access to all of the data in the study and accept responsibility for the decision to submit for publication.

Additional files

Supplementary file 1. *The proportion of different B cell subpopulations between healthy controls and acute HFRS patients.* [↗](#)

Supplementary file 2. *Difference in the main IgG glycome features of HFRS patients between age groups.* [↗](#)

Supplementary file 3. *Difference in the main IgG glycome features of HFRS patients between sex groups.* [↗](#)

Supplementary file 4. *Differentially expressed genes of different B cell subgroups between healthy controls and acute HFRS patients.* [↗](#)

Supplementary file 5. *Differentially expressed genes of different B cell subgroups a between health and acute HFRS groups* [↗](#)

Supplementary file 6. *Differential expression genes in the ASM, PB and RM B cell subpopulations.* [↗](#)

Supplementary file 7. *Differential expression genes in the DN, IM and naive B cell subpopulations.* [↗](#)

References

- Abu-Raya B, Michalski C, Sadarangani M, Lavoie PM (2020) **Maternal immunological adaptation during normal pregnancy** *Front Immunol* **11**:575197 <https://doi.org/10.3389/fimmu.2020.575197> | Google Scholar
- Ackerman ME, Das J, Pittala S, Broge T, Linde C, Suscovich TJ, Brown EP, Bradley T, Natarajan H, Lin S, Sassic JK, O'Keefe S, Mehta N, Goodman D, Sips M, Weiner JA, Tomaras GD, Haynes BF, Lauffenburger DA, Bailey-Kellogg C, Roederer M, Alter G. (2018) **Route of immunization defines multiple mechanisms of vaccine-mediated protection against SIV** *Nat Med* **24**:1590–1598 <https://doi.org/10.1038/s41591-018-0161-0> | Google Scholar
- Chakraborty S, Gonzalez JC, Sievers BL, Mallajosyula V, Chakraborty S, Dubey M, Ashraf U, Cheng BY, Kathale N, Tran KQT, Scallan C, Sinnott A, Cassidy A, Chen ST, Gelbart T, Gao F, Golan Y, Ji X, Kim-Schulze S, Prah M, Gaw SL, Gnjatich S, Marron TU, Merad M, Arunachalam PS, Boyd SD, Davis MM, Holubar M, Khosla C, Maecker HT, Maldonado Y, Mellins ED, Nadeau KC, Pulendran B, Singh U, Subramanian A, Utz PJ, Sherwood R, Zhang S, Jagannathan P, Tan GS, Wang TT (2022) **Early non-neutralizing, afucosylated antibody responses are associated with COVID-19 severity** *Sci Transl Med* **14**:eabm7853 <https://doi.org/10.1126/scitranslmed.abm7853> | Google Scholar
- García M, Iglesias A, Landoni VI, Bellomo C, Bruno A, Córdoba MT, Balboa L, Fernández GC, Sasiain MD, Martínez VP, Schierloh P. (2017) **Massive plasmablast response elicited in the acute phase of hantavirus pulmonary syndrome** *Immunology* **151**:122–135 <https://doi.org/10.1111/imm.12713> | Google Scholar
- Giron LB, Colomb F, Papasavvas E, Azzoni L, Yin X, Fair M, Anzurez A, Damra M, Mounzer K, Kostman JR, Tebas P, O'Doherty U, Tateno H, Liu Q, Betts MR, Montaner LJ, Abdel-Mohsen M. (2020) **Interferon-α alters host glycosylation machinery during treated HIV infection** *EBioMedicine* **59**:102945 <https://doi.org/10.1016/j.ebiom.2020.102945> | Google Scholar
- Hou H, Xu X, Sun F, Zhang X, Dong H, Wang L, Ge S, An K, Sun Q, Li Y, Cao W, Song M, Hu S, Xing W, Wang W, Li D, Wang Y. (2019) **Hyperuricemia is associated with immunoglobulin G N-glycosylation: A community-based study of glycan biomarkers** *Omic* **23**:660–667 <https://doi.org/10.1089/omi.2019.0004> | Google Scholar
- Hou H, Yang H, Liu P, Huang C, Wang M, Li Y, Zhu M, Wang J, Xu Y, Wang Y, Ma Q, Li D, Liao P, Wang W. (2021) **Profile of immunoglobulin G N-glycome in COVID-19 patients: A case-control study** *Front Immunol* **12**:748566 <https://doi.org/10.3389/fimmu.2021.748566> | Google Scholar
- Houde D, Peng Y, Berkowitz SA, Engen JR (2010) **Post-translational modifications differentially affect IgG1 conformation and receptor binding** *Mol Cell Proteomics* **9**:1716–1728 <https://doi.org/10.1074/mcp.M900540-MCP200> | Google Scholar
- Irvine EB, Alter G. (2020) **Understanding the role of antibody glycosylation through the lens of severe viral and bacterial diseases** *Glycobiology* **30**:241–253 <https://doi.org/10.1093/glycob/cwaa018> | Google Scholar

Jennewein MF, Alter G. (2017) **The immunoregulatory roles of antibody glycosylation** *Trends Immunol* **38**:358–372 <https://doi.org/10.1016/j.it.2017.02.004> | Google Scholar

Jin C, Chen Y, Wang Y, Li J, Liang J, Zheng S, Zhang L, Li Q, Wang Y, Ling F, Li Y, Zheng Y, Nie Q, Feng Q, Wang J, Yang H. (2024) **Single-cell RNA sequencing reveals special basal cells and fibroblasts in idiopathic pulmonary fibrosis** *Sci Rep* **14**:15778 <https://doi.org/10.1038/s41598-024-66947-5> | Google Scholar

Kerkman PF, Dernstedt A, Tadala L, Mittler E, Dannborg M, Sundling C, Maleki KT, Tauriainen J, Tuiskunen-Bäck A, Wigren Byström J, Ocaya P, Thunberg T, Jangra RK, Román-Sosa G, Guardado-Calvo P, Rey FA, Klingström J, Chandran K, Puhar A, Ahlm C, Forsell MN (2021) **Generation of plasma cells and CD27-IgD- B cells during hantavirus infection is associated with distinct pathological findings** *Clin Transl Immunology* **10**:e1313 <https://doi.org/10.1002/cti2.1313> | Google Scholar

Larsen MD, de Graaf EL, Sonneveld ME, Plomp HR, Nouta J, Hoepel W, Chen HJ, Linty F, Visser R, Brinkhaus M, ušić T, de Taeye SW, Bentlage AEH, Toivonen S, Koeleman CAM, Sainio S, Kootstra NA, Brouwer PJM, Geyer CE, Derksen NIL, Wolbink G, de Winther M, Sanders RW, van Gils MJ, de Bruin S, Vlaar APJ, Rispens T, den Dunnen J, Zaaijer HL, Wuhrer M, van der Schoot CE, Vidarsson G. (2021) **Afucosylated IgG characterizes enveloped viral responses and correlates with COVID-19 severity** *Science* **371** <https://doi.org/10.1126/science.abc8378> | Google Scholar

Li R, Sun J, Chen Y, Fan X, Wang X, Zhang X, Zhang K, Han Q, Liu Z. (2023) **Clinical and laboratory features and factors predicting disease severity in pediatric patients with hemorrhagic fever with renal syndrome caused by Hantaan virus** *J Med Virol* **95**:e28339 <https://doi.org/10.1002/jmv.28339> | Google Scholar

Li Y, Quan C, Xing W, Wang P, Gao J, Zhang Z, Jiang X, Ma C, Carr MJ, He Q, Gao L, Bi Y, Tang H, Shi W. (2020) **Rapid humoral immune responses are required for recovery from haemorrhagic fever with renal syndrome patients** *Emerg Microbes Infect* **9**:2303–2314 <https://doi.org/10.1080/22221751.2020.1830717> | Google Scholar

Liechti T, Roederer M. (2019) **OMIP-051-28-color flow cytometry panel to characterize B cells and myeloid cells** *Cytometry A* **95**:150–155 <https://doi.org/10.1002/cyto.a.23689> | Google Scholar

Liu D, Chu X, Wang H, Dong J, Ge SQ, Zhao ZY, Peng HL, Sun M, Wu LJ, Song MS, Guo XH, Meng Q, Wang YX, Lauc G, Wang W. (2018) **The changes of immunoglobulin G N-glycosylation in blood lipids and dyslipidaemia** *J Transl Med* **16**:235 <https://doi.org/10.1186/s12967-018-1616-2> | Google Scholar

Liu P, Wang X, Dun A, Li Y, Li H, Wang L, Zhang Y, Li C, Zhang J, Zhang X, Ma L, Hou H. (2023) **High-throughput profiling of serological immunoglobulin G N-glycome as a noninvasive biomarker of gastrointestinal cancers** *Engineering* **26**:44–53 <https://doi.org/10.1016/j.eng.2023.02.008> | Google Scholar

Lu LL, Suscovich TJ, Fortune SM, Alter G. (2018) **Beyond binding: antibody effector functions in infectious diseases** *Nat Rev Immunol* **18**:46–61 <https://doi.org/10.1038/nri.2017.106> | Google Scholar

Ma Y, Yuan B, Zhuang R, Zhang Y, Liu B, Zhang C, Zhang Y, Yu H, Yi J, Yang A, Jin B. (2015) **Hantaan virus infection induces both Th1 and ThGranzyme B+ cell immune responses**

that associated with viral control and clinical outcome in humans *PLoS Pathog* **11**:e1004788 <https://doi.org/10.1371/journal.ppat.1004788> | [Google Scholar](#)

Pereira MS, Durães C, Catarino TA, Costa JL, Cleyne I, Novokmet M, Krištić J, tambuk J, Conceição-Neto N, Machado JC, Marcos-Pinto R, Magro F, Vermeire S, Lauc G, Lago P, Pinho SS (2020) **Genetic variants of the MGAT5 gene are functionally implicated in the modulation of T cells glycosylation and plasma IgG glycome composition in ulcerative colitis** *Clin Transl Gastroenterol* **11**:e00166 <https://doi.org/10.14309/ctg.000000000000166> | [Google Scholar](#)

Song P, Zheng N, Liu Y, Tian C, Wu X, Ma X, Chen D, Zou X, Wang G, Wang H, Zhang Y, Lu S, Wu C, Wu Z. (2018) **Deficient humoral responses and disrupted B-cell immunity are associated with fatal SFTSV infection** *Nat Commun* **9**:3328 <https://doi.org/10.1038/s41467-018-05746-9> | [Google Scholar](#)

Tang K, Zhang C, Zhang Y, Zhang Y, Du H, Jin B, Ma Y. (2019) **Elevated plasma interleukin 34 levels correlate with disease severity-reflecting parameters of patients with haemorrhagic fever with renal syndrome** *Infect Dis (Lond)* **51**:847–853 <https://doi.org/10.1080/23744235.2019.1672887> | [Google Scholar](#)

Trastoy B, Klontz E, Orwenyo J, Marina A, Wang LX, Sundberg EJ, Guerin ME (2018) **Structural basis for the recognition of complex-type N-glycans by Endoglycosidase S** *Nat Commun* **9**:1874 <https://doi.org/10.1038/s41467-018-04300-x> | [Google Scholar](#)

Wang TT, Sewatanon J, Memoli MJ, Wrammert J, Bournazos S, Bhaumik SK, Pinsky BA, Chokephaibulkit K, Onlamoon N, Pattanapanyasat K, Taubenberger JK, Ahmed R, Ravetch JV (2017) **IgG antibodies to dengue enhanced for FcγRIIIA binding determine disease severity** *Science* **355**:395–398 <https://doi.org/10.1126/science.aai8128> | [Google Scholar](#)

Wieczorek M, Braicu EI, Oliveira-Ferrer L, Sehouli J, Blanchard V. (2020) **Immunoglobulin G subclass-specific glycosylation changes in primary epithelial ovarian cancer** *Front Immunol* **11**:654 <https://doi.org/10.3389/fimmu.2020.00654> | [Google Scholar](#)

Zhang H, Wang Y, Ma Y, Tang K, Zhang C, Wang M, Zhang X, Xue M, Jia X, Hu H, Li N, Zhuang R, Jin B, Chen L, Zhang Y, Zhang Y. (2022) **Increased CD4(+)CD8(+) double positive T cells during Hantaan virus infection** *Viruses* **14**:2243 <https://doi.org/10.3390/v14102243> | [Google Scholar](#)

Zheng Z, Wang P, Wang Z, Zhang D, Wang X, Zuo S, Li X. (2019) **The characteristics of current natural foci of hemorrhagic fever with renal syndrome in Shandong Province, China, 2012-2015** *PLoS Negl Trop Dis* **13**:e0007148 <https://doi.org/10.1371/journal.pntd.0007148> | [Google Scholar](#)

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Reviewer #1 (Public review):

Summary:

The authors investigated the potential role of IgG N-glycosylation in Haemorrhagic Fever with Renal Syndrome (HFRS), which may offer significant insights for understanding molecular mechanisms and for the development of therapeutic strategies for this infectious disease. However, several issues need to be addressed.

Major Points:

(1) The authors should provide a detailed description of the pathogenesis of Haemorrhagic Fever with Renal Syndrome (HFRS) and elaborate on the crucial role of IgG proteins in the disease's progression (line 65).

(2) An additional discussion on the significance of glycosylation, particularly IgG N-glycosylation, in viral infections should be included in the Introduction section.

(3) In the Abstract section, the authors state that HTNV-specific IgG antibody titers were detected and IgG N-glycosylation was analyzed. However, the analysis of plasma IgG N-glycans is described in the Methods section. Therefore, the authors should clarify the glycome analysis process. Was the specific IgG glycome profile similar to the total IgG N-glycome? Given the biological relevance of specific IgG in immunological diseases, characterizing the specific IgG N-glycome profile would be more significant than analyzing the total plasma IgG.

(4) Further details regarding the N-glycome analysis should be provided, including the quantity of IgG protein used and the methodology employed for analyzing IgG N-glycans (lines 286-287).

(5) Additional statistical analyses should be performed, including multiple comparisons with p-value adjustment, false discovery rate (FDR) control, and Pearson correlation (line 291).

(6) Quality control should be conducted prior to the IgG N-glycome analysis. Additionally, both biological and technical replicates are essential to assess the reproducibility and robustness of the methods.

(7) Multiple regression analysis should be conducted to evaluate the influence of genetic and environmental factors on the IgG N-glycome.

(8) Line 196. Additional discussions should be included, focusing on the underlying correlation between the differential expression of B-cell glycogenes and the dysregulated IgG N-glycome profile, as well as the potential molecular mechanisms of IgG N-glycosylation in the development of HFRS.

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Reviewer #2 (Public review):

Summary:

This work sought to explore antibody responses in the context of hemorrhagic fever with renal syndrome (HFRS) - a severe disease caused by Hantaan virus infection. Little is known about the characteristics or functional relevance of IgG Fc glycosylation in HFRS. To address this gap, the authors analyzed samples from 65 patients with HFRS spanning the acute and convalescent phases of disease via IgG Fc glycan analysis, scRNAseq, and flow cytometry. The authors observed changes in Fc glycosylation (increased fucosylation and decreased bisection) coinciding with a 4-fold or greater increase in Hantaan virus-specific antibody titer. They suggest that these shifts contribute to disease recovery. The study also includes exploratory analyses linking IgG glycan profiles to glycosylation-related gene expression in distinct B cell subsets, using single-cell transcriptomics. Overall, this is an interesting study that combines serological profiling with transcriptomic data to shed light on humoral immune responses in an underexplored infectious disease. The integration of Fc glycosylation data with single-cell transcriptomic data is a strength. However, some improvements could be made in the clarity of both the Results and Materials and Methods sections, and some conclusions would benefit from greater caution, particularly in avoiding overinterpretation of correlative findings.

Comments:

(1) While it is great to reference prior publications in the Materials and Methods section, the current level of detail is insufficient to clearly understand the study design and experimental procedures performed. Readers should not be expected to consult multiple previous papers to grasp the core methodological aspects of the present paper. For instance, the categorization

of HFRS patients into different clinical subtypes/courses, and the methods for measuring Fc glycosylation should be explicitly described in the Materials and Methods section of this manuscript.

(2) The authors should explain the nature of their cohort in a bit more detail. While it appears that HFRS cases were identified based on IgM ELISA and/or PCR, these are indicators of the Haantan virus infection. My understanding is that not all Haantan virus infections progress to HFRS. Thus, it is unclear whether all patients in the HFRS group actually had hemorrhagic fever. This distinction is critical for interpreting how the results observed relate to disease severity.

(3) The authors state that: "A 4-fold or greater increase in HTNV-NP-specific antibody titers usually indicates a protective humoral immune response during the acute phase", but they do not cite any references or provide any context that supports this claim. Given that in their own words, one of the most significant findings in the study is changes in glycosylation coinciding with this 4-fold increase, it is important to ground this claim in evidence. Without this, the use of a 4-fold threshold appears arbitrary and weakens the rationale for using this immune state as a proxy for protective immunity.

(4) The authors also claim that changes in Fc glycosylation influence recovery from HFRS - a point even emphasized in the manuscript title. However, this conclusion is not well supported by the data for two main reasons. First, the authors appear to measure bulk IgG Fc glycans, not Fc glycans of Hantaan virus-specific antibodies. While reasonable, this is something that should be communicated in the manuscript. Hantaan virus-specific antibodies are likely a very small fraction of total circulating IgG antibodies (perhaps ~1%), even during acute infection. As a result, changes in bulk Fc glycosylation may (or may not) accurately reflect the glycosylation state of Hantaan virus-specific antibodies. Second, even if the bulk Fc glycan shifts do mirror those of Hantaan virus-specific antibodies, it remains unclear whether these changes causally drive recovery or are merely a consequence of the infection being resolved. Thus, while the differences in Fc glycosylation observed are interesting - and it is tempting to speculate on their functional significance - the manuscript treats the observed correlations as causal mechanistic insight without sufficient data or justification.

(5) Fc glycosylation is known to be influenced by covariates such as age and sex. While it is helpful that the authors stratified the patients by age group and looked for significant differences in glycosylation across them, a more robust approach would be to directly control for these covariates in the statistical analysis - such as by using a linear mixed effects model, in which disease state (e.g., acute vs. convalescent), age, and sex are treated as fixed effects, and subject ID is included as a random effect to account for repeated measures. This would allow the authors to assess whether observed differences in Fc glycosylation remain significant after accounting for potential confounders. This could be important given that some of the reported differences are quite small, for example, 94.29% vs. 94.89% fucosylation.

(6) The manuscript states that there are limited studies on antibody glycosylation in the context of HFRS, but does not cite any relevant literature. If prior work exists, it should be cited to contextualize the current study. If no prior studies have been conducted/reported, to the author's knowledge, that should be stated explicitly to show the novelty of the work.

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